

TITLE OF THE PROJECT

Simple patient health monitoring system

Abstract

The Health Monitoring system is designed to continuously track and record vital health parameters, and blood oxygen levels using integrated sensors and microcontrollers. The system aims to provide real-time health data to users and healthcare professionals through a user-friendly interface, enabling early detection of abnormal conditions and facilitating prompt medical intervention. The data collected can be transmitted wirelessly to mobile devices or cloud servers, ensuring remote accessibility and continuous monitoring. This project not only enhances patient care but also reduces the burden on healthcare infrastructure by enabling at-home health tracking and emergency alert systems.

Components Utilized

- Arduino UNO
- Breadboard(Mini)
- LCD Display(20*2)
- MAX30100 Sensor
- Jumper wires
- Power Supply

ARDUINO UNO

Arduino UNO is a low-cost, flexible, and easy-to-use programmable open-source microcontroller board that can be integrated into a variety of electronic projects. This board can be interfaced with other Arduino boards, Arduino shields, Raspberry Pi boards and can control relays, LEDs, servos, and motors as an output.

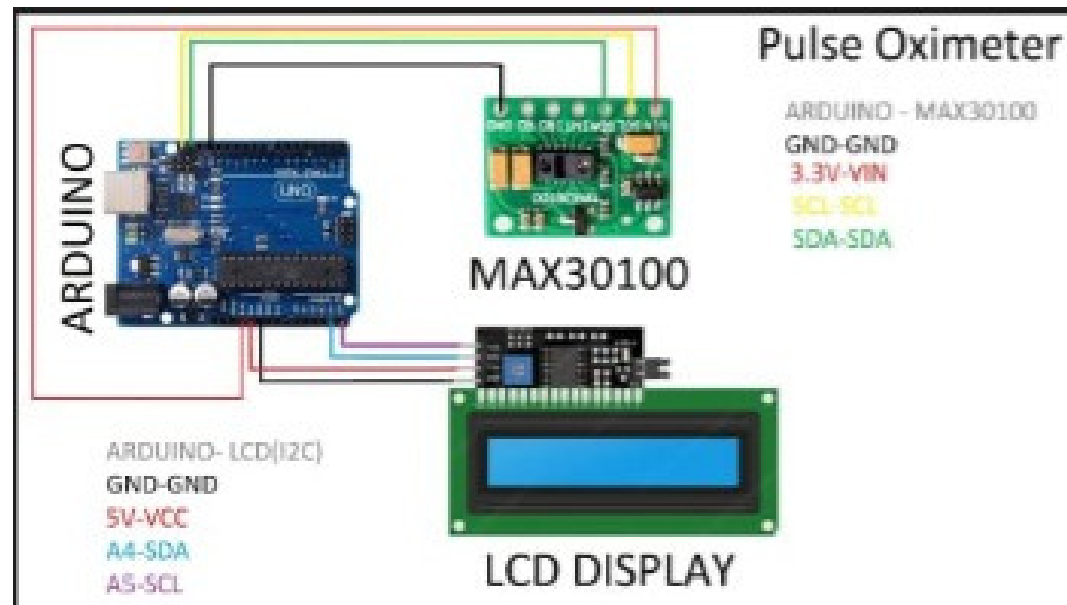


MAX30100 SENSOR

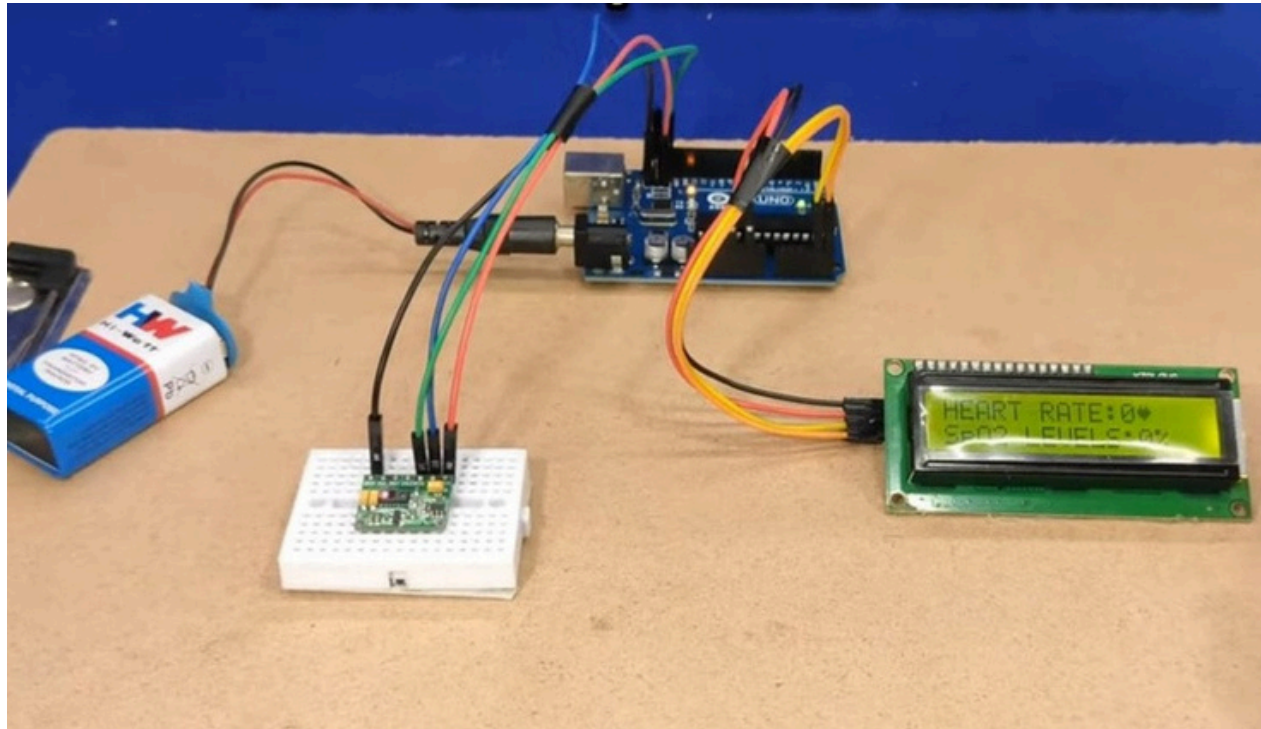
The MAX30100 pulse oximetry & heart-rate monitor sensor is a low-power, I2C-based plug & play biometric sensor for detecting pulse oximetry & heart-rate signals. This sensor includes two LEDs like RED & IR LED, a photodetector, modifiable optics & low noise signal processor to detect heart pulse rate signals. On the right side of this sensor, two LEDs are available whereas on the left side, it has a sensitive photodetector. This sensor is used in various medical & industrial equipment like fitness measurement devices, wearable instruments & medical devices.



CIRCUIT DIAGRAM



CIRCUIT CONNECTIONS



WORKING

A simple patient health monitoring system works by continuously collecting vital health parameters using sensors such as heart rate monitors, blood oxygen monitor. A microcontroller, like an Arduino or ESP8266, processes this data through max30100 oximeter sensor and gives the output on LCD Display . This whole data will be useful for the doctor to analyze patient health and also can be used in home by ourselves

CODE

```
#include <Wire.h>
#include "MAX30100_PulseOximeter.h"
#include <LiquidCrystal_I2C.h>
#define REPORTING_PERIOD_MS
    1000
```

```
LiquidCrystal_I2C lcd(0x27,16,2);
```

```
PulseOximeter pox;
uint32_t tsLastReport = 0;
    int sp = 0;
    int hr = 0;
```

```
byte Heart[] = {
    B00000,
    B01010,
    B11111,
    B11111,
    B01110,
    B00100,
    B00000,
    B00000
};
```

```
void onBeatDetected() {  
    Serial.println("Beat!");  
    digitalWrite(13,HIGH);  
    digitalWrite(5,HIGH); delay(10);  
    digitalWrite(13,LOW);  
    digitalWrite(5,LOW); } void setup() {  
    Serial.begin(9600);  
    Serial.print("Initializing pulse  
oximeter.."); pinMode(13, OUTPUT);  
    pinMode(5, OUTPUT); lcd.init();  
    lcd.backlight(); lcd.createChar(0,  
        Heart);
```

```
    lcd.setCursor(0,0  
);
```

```
    lcd.print(" PATIENT  
        HEALTH ");  
    lcd.setCursor(0,1);  
    lcd.print(" MONITORING  
"); delay(2000); lcd.clear();  
    lcd.setCursor(0,0);  
    lcd.print(" SYSTEM USING  
"); lcd.setCursor(0,1);  
    lcd.print(" ARDUINO ");  
    delay(2000); lcd.clear();
```

```
    if (!pox.begin()) {  
        Serial.println("FAILED");
```

```
        for(;;);  
    } else {  
        Serial.println("SUCCESS");
```

```
    }
```

```
pox.setIRLedCurrent(MAX30100_LED_CURR_30_6MA);  
pox.setOnBeatDetectedCallback(onBeatDetected);  
}
```

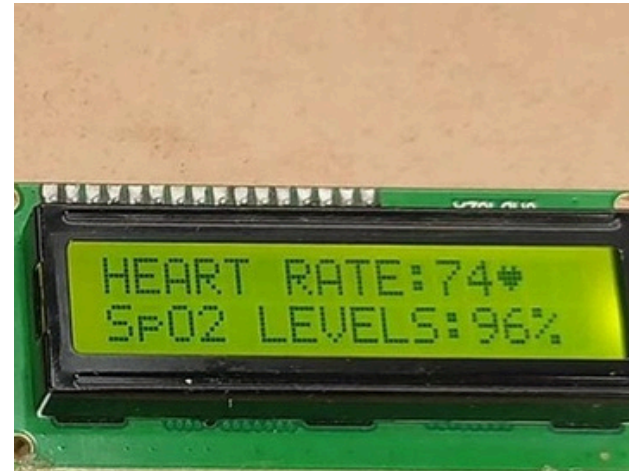
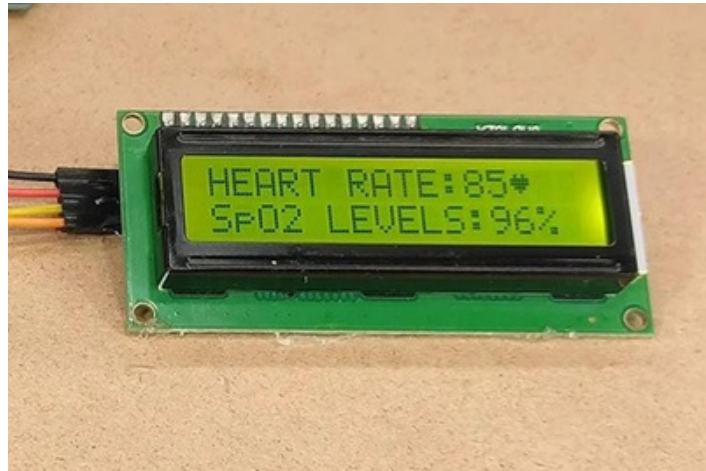
```
void loop() {  
    pox.update();  
    if (millis() - tsLastReport > REPORTING_PERIOD_MS)  
    {  
        hr = pox.getHeartRate();  
        sp = pox.getSpO2();
```

```
    lcd.setCursor(0,0);  
    lcd.print("HEARTRATE:");  
        .print(hr);  
        lcd.write(0);  
        lcd.print("");  
Serial.print("HEARTRATE:");  
        l.print(hr);
```

```
    lcd.setCursor(0,1);  
    lcd.print("SpO2 LEVELS:");  
        lcd.print(sp);  
        lcd.print (" % ");  
Serial.print("bpm / SpO2:");  
    Serial.print(sp);  
    Serial.println("%");
```

```
tsLastReport = millis();  
    }  
}
```

SIMULATION RESULTS



Advantages

Early Detection: It helps identify potential health issues before they become severe, allowing timely intervention.

Chronic Disease Management: Patients with conditions like diabetes or hypertension can track their health in real-time, improving management.

Remote Monitoring: Patients can be monitored outside hospital settings, reducing the need for frequent visits and improving accessibility.

Applications

Post-Surgery Recovery: Helps patients track their recovery progress and ensures timely medical attention if complications arise.

Fitness & Wellness Tracking: Used by individuals to monitor heart rate, oxygen levels, and other health metrics for personal wellness.

IoT-Based Healthcare: Integrated with IoT platforms like ThingSpeak, allowing real-time data transmission and analysis.

Emergency Alerts: Systems can detect abnormal health patterns and send alerts for immediate medical intervention.

Conclusion

A simple patient health monitoring system plays a crucial role in modern healthcare by enabling continuous tracking of vital signs, early detection of health issues, and improved accessibility to medical care. By leveraging technologies like IoT and microcontrollers, these systems enhance patient safety, reduce hospital visits, and facilitate remote monitoring, making healthcare more efficient and cost-effective

Furture Scope

AI and Predictive Analytics: AI-driven algorithms analyze patient data in real time, identifying patterns and predicting health deterioration before symptoms appear.

Wearable Devices and Sensors: Smartwatches and biosensors continuously track vital signs like heart rate, blood pressure, and glucose levels, enabling proactive healthcare.

Remote Patient Monitoring (RPM): Telehealth and cloud-based platforms allow doctors to monitor patients remotely, reducing hospital visits and improving care for chronic conditions.

Multi-Parameter Monitoring: Next-generation patient monitors integrate multiple health metrics, such as ECG tracking and oxygen saturation, for comprehensive health assessments

References

Health Monitoring Systems: An Enabling Technology for Patient Care – This book provides a comprehensive overview of remote health monitoring using wearable sensors, IoT- based systems, and biomedical signal processing.

Real-Time Patient Health Data Monitoring System Based on IoT Technology – A research-based book discussing IoT applications in patient health monitoring, including real-time data collection and analysis.

THANK YOU